

1. 二重积分

1.3、二重积分的计算

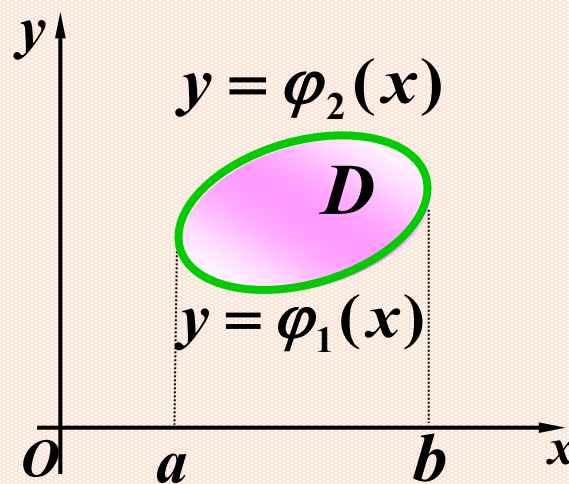
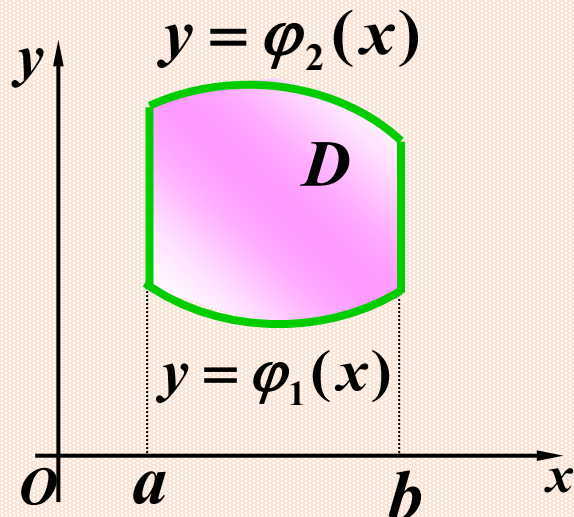
本节介绍计算二重积分的方法:
二重积分化为累次积分(即两次定积分).

1.3 利用直角坐标系计算二重积分

(1) 积分区域为: $a \leq x \leq b$, $\varphi_1(x) \leq y \leq \varphi_2(x)$.

其中函数 $\varphi_1(x)$ 、 $\varphi_2(x)$ 在区间 $[a, b]$ 上连续.

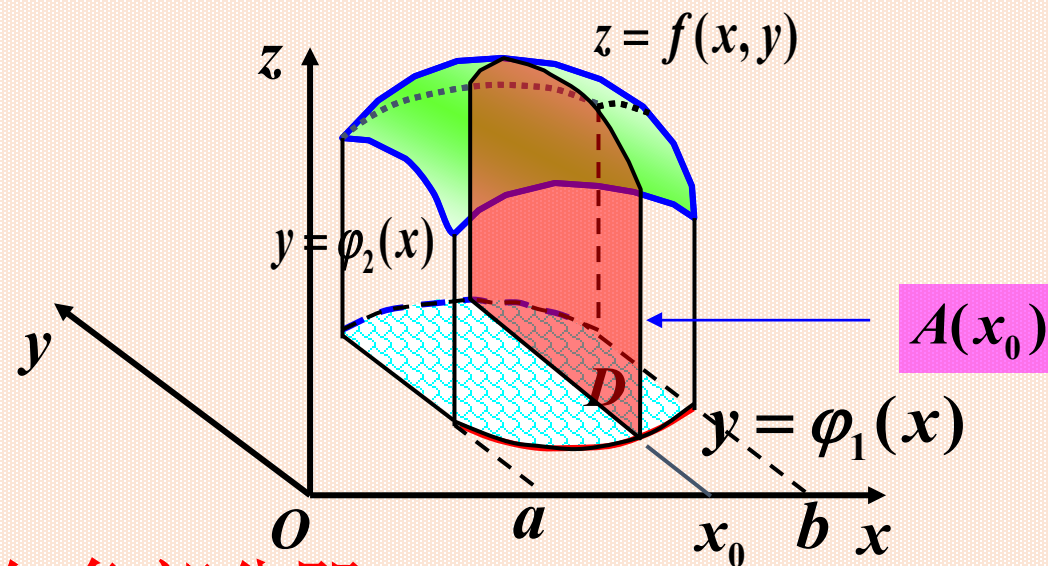
X-型



用二重积分的几何意义说明其算法

$\therefore \iint_D f(x, y) d\sigma$ ($f(x, y) \geq 0$) 的值等于以 D 为底, 以曲面 $z = f(x, y)$ 为顶的曲顶柱体的体积.

应用计算“平行截面面积为已知的立体求体积”的方法.



* 计算截面面积 (红色部分即 $A(x_0)$)

是区间 $[\varphi_1(x_0), \varphi_2(x_0)]$ 为底, 曲线 $z = f(x_0, y)$ 为曲边的曲边梯形.

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曲线 $z = f(x_0, y)$ 为曲边
的曲边梯形.

$$A(x_0) = \int_{\varphi_1(x_0)}^{\varphi_2(x_0)} f(x_0, y) dy$$

$\forall x \in [a, b]$ 有:

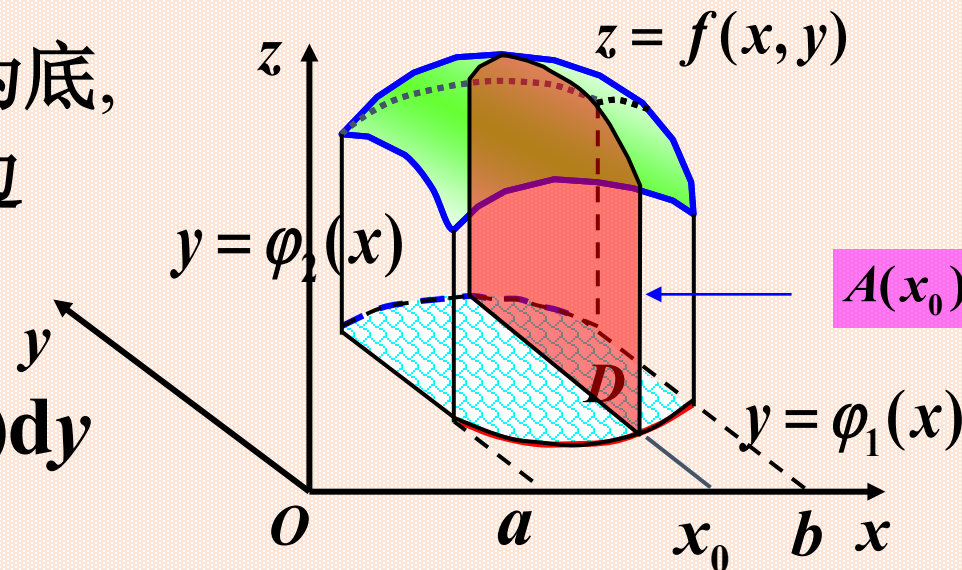
$$A(x) = \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy$$

$$\iint_D f(x, y) d\sigma = \int_a^b dx \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy$$

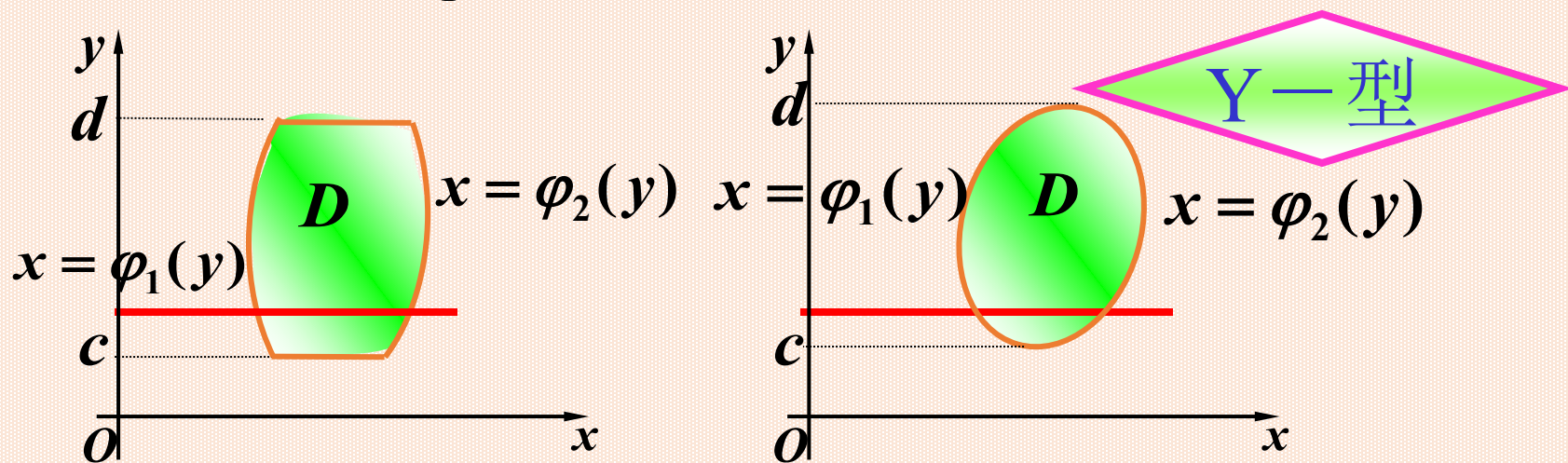
$$= \int_a^b \left(\int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy \right) dx$$

称为累次积分.

先对 y 后对 x 的二次积分



(2) 积分区域为: $c \leq y \leq d$, $\varphi_1(y) \leq x \leq \varphi_2(y)$
 其中函数 $\varphi_1(y)$ 、 $\varphi_2(y)$ 在区间 $[c, d]$ 上连续.

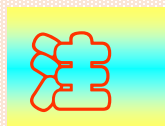


$$\iint_D f(x, y) d\sigma = \int_c^d \left(\int_{\varphi_1(y)}^{\varphi_2(y)} f(x, y) dx \right) dy$$

先对 x 后对 y 的二次积分

也即

$$\iint_D f(x, y) d\sigma = \int_c^d dy \int_{\varphi_1(y)}^{\varphi_2(y)} f(x, y) dx$$



特殊地 D 为矩形域: $a \leq x \leq b, c \leq y \leq d$

$$\begin{aligned}\text{则 } \iint_D f(x, y) d\sigma &= \int_a^b dx \int_c^d f(x, y) dy \\ &= \int_c^d dy \int_a^b f(x, y) dx\end{aligned}$$

如 D 是上述矩形域, 且 $f(x, y) = f_1(x) \cdot f_2(y)$

$$\begin{aligned}\text{则 } \iint_D f_1(x) f_2(y) dx dy &= \int_a^b \left(\int_c^d f_1(x) \cdot f_2(y) dy \right) dx \\ &= \int_a^b \left(f_1(x) \cdot \int_c^d f_2(y) dy \right) dx \\ \text{得 } &= \int_a^b f_1(x) dx \int_c^d f_2(y) dy\end{aligned}$$

即等于两个定积分的乘积.

X型区域的特点: 穿过区域且平行于y轴的直线与区域边界相交不多于两个交点.

Y型区域的特点: 穿过区域且平行于x轴的直线与区域边界相交不多于两个交点.

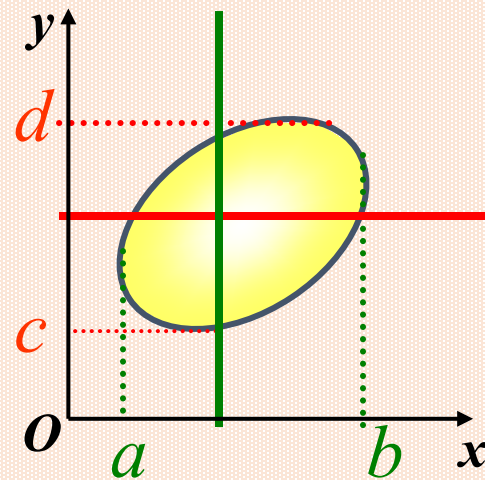
(3)积分区域D既是X型:

$$a \leq x \leq b, \quad \varphi_1(x) \leq y \leq \varphi_2(x)$$

又是Y型:

$$c \leq y \leq d, \quad \varphi_1(y) \leq x \leq \varphi_2(y)$$

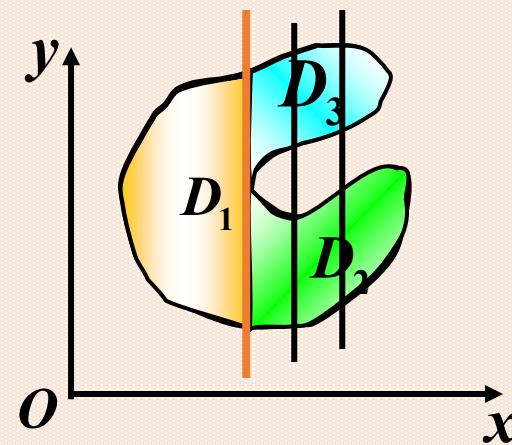
计算结果一样. 但可作出适当选择.



(4) 若区域如图, 则必须分割.

在分割后的三个区域上分别使用积分公式.

(用积分区域的可加性质)



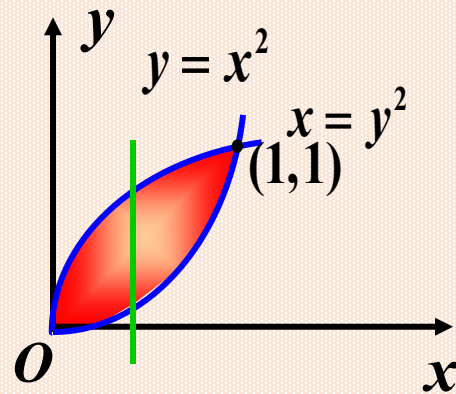
$$\iint_D = \iint_{D_1} + \iint_{D_2} + \iint_{D_3}$$

D_1 、 D_2 、 D_3 都是X型区域

例1 求 $\iint_D (x^2 + y) dx dy$, 其中 D 是抛物线 $y = x^2$ 和 $x = y^2$ 所围平面闭区域.

解 两曲线的交点

$$\begin{cases} x = y^2 \\ y = x^2 \end{cases} \Rightarrow (1,1), (0,0)$$



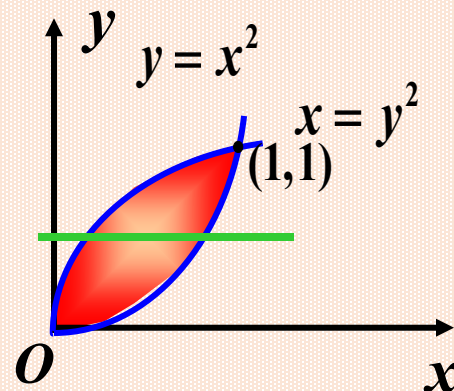
积分域既是X型又是Y型

法一

$$\begin{aligned} \iint_D (x^2 + y) dx dy &= \int_0^1 dx \int_{x^2}^{\sqrt{x}} (x^2 + y) dy \\ &= \int_0^1 [x^2(\sqrt{x} - x^2) + \frac{1}{2}(x - x^4)] dx = \frac{33}{140}. \end{aligned}$$

$$\iint_D (x^2 + y) dx dy$$

法二 先对 x 后对 y 的积分?



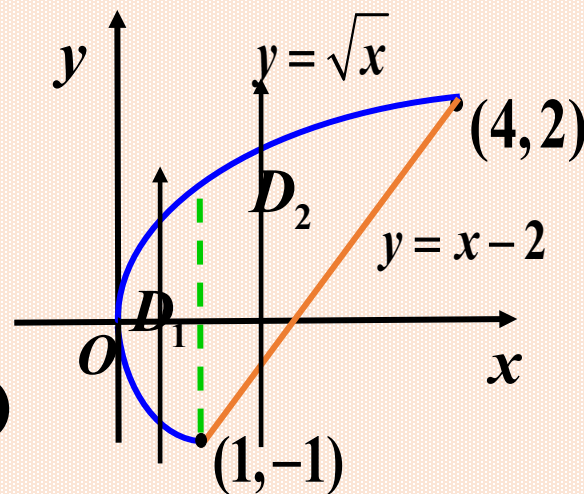
$$\begin{aligned} \iint_D (x^2 + y) dx dy &= \int_0^1 dy \int_{y^2}^{\sqrt{y}} (x^2 + y) dx \\ &= \frac{33}{140} \end{aligned}$$

可以看出：这题中先对 x 后对 y 积分和先对 y 后对 x 的积分都可以得出结果，计算难度一样。

例2 求 $\iint_D xy dx dy$, 其中 D 是抛物线 $y^2 = x$ 和直线 $y = x - 2$ 所围平面闭区域.

解 两曲线的交点

$$\begin{cases} y^2 = x \\ y = x - 2 \end{cases} \Rightarrow (4, 2), (1, -1)$$



法一 $D = D_1 \cup D_2$,

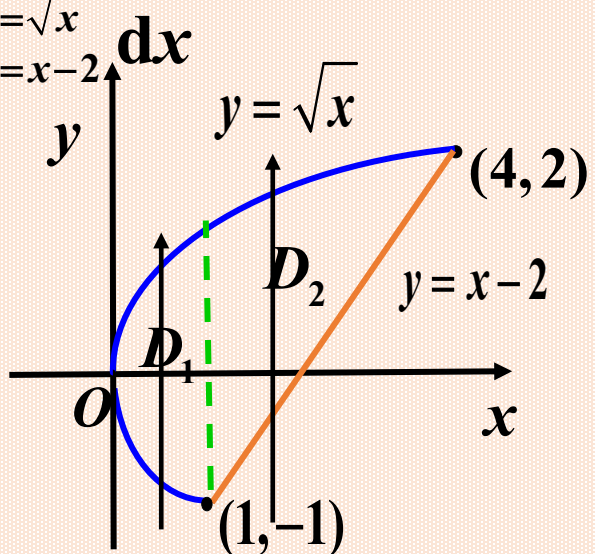
$$D_1 = \{(x, y) \mid 0 \leq x \leq 1, -\sqrt{x} \leq y \leq \sqrt{x}\}$$

$$D_2 = \{(x, y) \mid 1 \leq x \leq 4, x - 2 \leq y \leq \sqrt{x}\}$$

注意到 D_1 关于 x 轴对称, 故由奇偶性, $\iint_{D_1} xy dx dy = 0$.

$$\begin{aligned}
 \iint_{D_2} xy dx dy &= \int_1^4 dx \int_{x-2}^{\sqrt{x}} xy dy = \int_1^4 x \left(\frac{y^2}{2} \right) \Big|_{y=x-2}^{y=\sqrt{x}} dx \\
 &= \frac{1}{2} \int_1^4 x [x - (x-2)^2] dx = \frac{45}{8}.
 \end{aligned}$$

故 $\iint_D xy dx dy = \iint_{D_1} xy dx dy + \iint_{D_2} xy dx dy = \frac{45}{8}.$

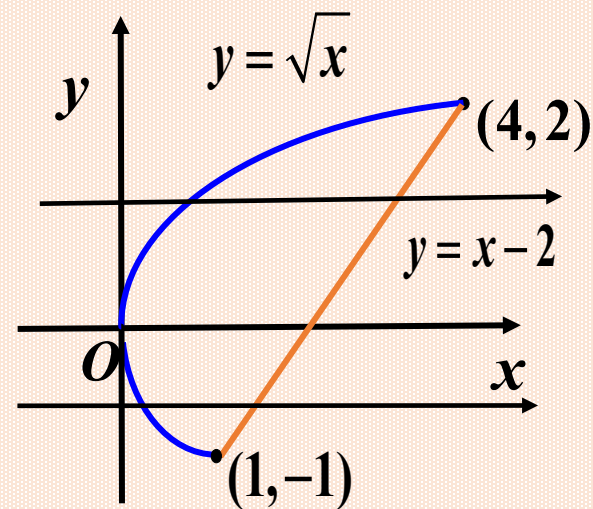


法二 把 D 看成 Y 型域

$$D = \{(x, y) \mid -1 \leq y \leq 2, y^2 \leq x \leq y + 2\}$$

故

$$\begin{aligned}
 \iint_D xy dx dy &= \int_{-1}^2 dy \int_{y^2}^{y+2} xy dx \\
 &= \frac{1}{2} \int_{-1}^4 y [(y+2)^2 - y^4] dy = \frac{45}{8}.
 \end{aligned}$$



例3 计算累次积分 $\int_0^1 dx \int_x^1 \sin y^2 dy$

分析 $\sin y^2$ 对 y 的积分不能用基本积分法算出，
而它对 x 的积分可用基本积分法算出。

所以将二次积分先交换积分次序。

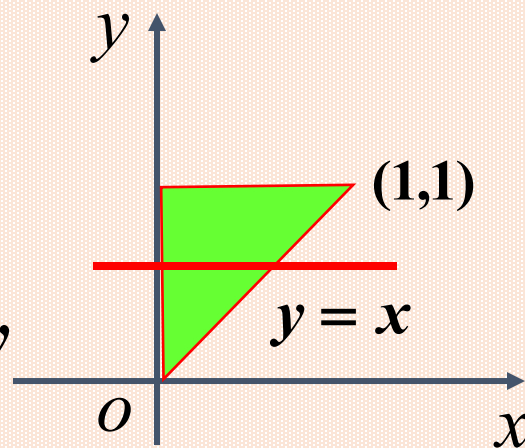
交换积分次序的方法是：

(1) 将所给的积分域用联立不等式表示 D ：

$$0 \leq x \leq 1, \quad x \leq y \leq 1$$

(2) 画出积分域的草图

(3) 改写 D 为： $0 \leq y \leq 1, 0 \leq x \leq y$



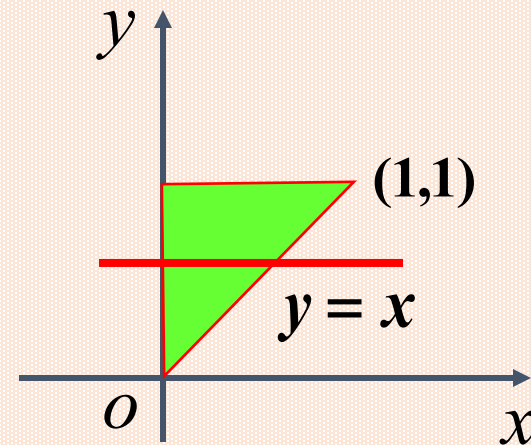
$$\int_0^1 dx \int_x^1 \sin y^2 dy = \iint_D \sin y^2 dx dy$$

$$= \int_0^1 dy \int_0^y \sin y^2 dx$$

$$= \int_0^1 (\sin y^2) \cdot x \Big|_0^y dy$$

$$= \int_0^1 y \sin y^2 dy$$

$$= \frac{1}{2} \int_0^1 \sin y^2 dy^2 = \frac{1}{2} (1 - \cos 1)$$



$$D : 0 \leq y \leq 1, \quad 0 \leq x \leq y$$

计算二重积分时, 恰当的选取积分次序十分重要, 它不仅涉及到计算繁简问题, 而且又是能否进行计算的问题.

若 D 既是 X 型域, 又是 Y 型域, 则计算形如 $\iint_D f(y)dx dy$ 的积分时, 一般按 Y 型域来算更简单.

特别是当 $\int f(y)dy$ 不好算的情形.

$$\iint_D f(y)dx dy = \int_c^d f(y)dy \int_{\phi_1(y)}^{\phi_2(y)} dx = \int_c^d f(y)[\phi_2(y) - \phi_1(y)]dy$$

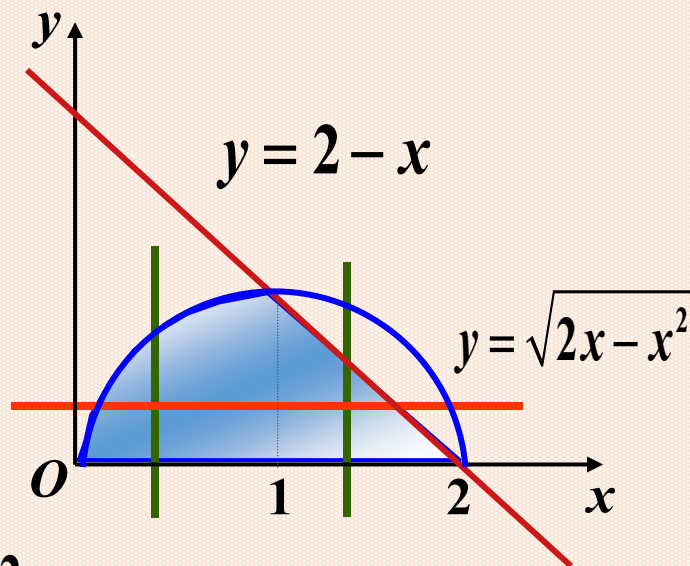
计算形如 $\iint_D f(x)dx dy$ 的积分则一般按 X 型域来算.

$$\iint_D f(x)dx dy = \int_a^b f(x)dx \int_{\varphi_1(x)}^{\varphi_2(x)} dy = \int_a^b f(x)[\varphi_2(x) - \varphi_1(x)]dx$$

例4 交换积分次序:

$$\int_0^1 dx \int_0^{\sqrt{2x-x^2}} f(x, y) dy + \int_1^2 dx \int_0^{2-x} f(x, y) dy$$

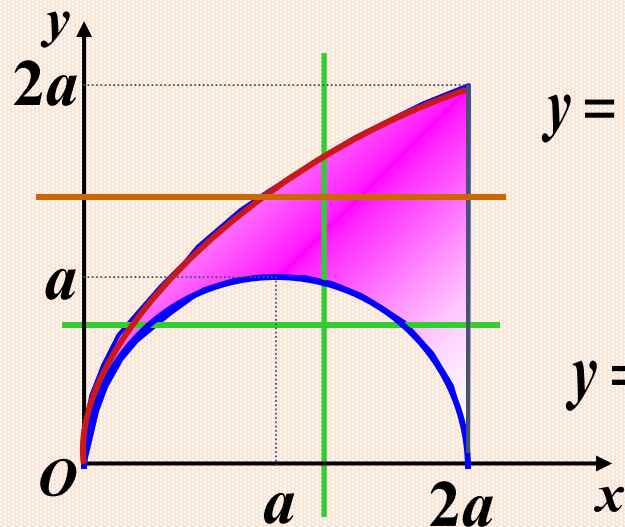
解 积分区域:



$$\text{原式} = \int_0^1 dy \int_{1-\sqrt{1-y^2}}^{2-y} f(x, y) dx$$

例5 交换积分次序: $\int_0^{2a} dx \int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x, y) dy \quad (a > 0)$

解



$$y = \sqrt{2ax} \Rightarrow x = \frac{y^2}{2a}$$

$$y = \sqrt{2ax - x^2} \Rightarrow x = a \pm \sqrt{a^2 - y^2}$$

$$\text{原式} = \int_0^a dy \int_{\frac{y^2}{2a}}^{a - \sqrt{a^2 - y^2}} f(x, y) dx$$

$$+ \int_0^a dy \int_{a + \sqrt{a^2 - y^2}}^{2a} f(x, y) dx + \int_a^{2a} dy \int_{\frac{y^2}{2a}}^{2a} f(x, y) dx$$

二重积分的计算规律(1)

交换积分次序的步骤

- (A) 将已给的二次积分的积分限得出相应的二重积分的积分区域，并画出草图；
- (B) 按相反顺序确定积分限，写出相应的二次积分.

二次积分一定能交换次序  答 不一定!

例如: 设 $f(x, y) = \begin{cases} \frac{x^2 - y^2}{(x^2 + y^2)^2}, & \text{当 } x^2 + y^2 \neq 0 \text{ 时,} \\ 0, & \text{当 } x^2 + y^2 = 0 \text{ 时.} \end{cases}$

$$I_1 = \int_0^1 dx \int_0^1 f(x, y) dy, \text{ 由于 } \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right) = f(x, y),$$

$$\text{故 } \int_0^1 f(x, y) dy = \int_0^1 \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right) dy = \left. \frac{y}{x^2 + y^2} \right|_0^1 = \frac{1}{x^2 + 1};$$

$$\begin{aligned} \text{所以 } I_1 &= \int_0^1 dx \int_0^1 f(x, y) dy = \int_0^1 \frac{1}{x^2 + 1} dx = \arctan x \Big|_0^1 \\ &= \frac{\pi}{4}, \end{aligned}$$

说明：当 $f(x, y)$ 在所考虑的区域上连续时，
二次积分可以交换积分次序。

$$I_1 = \int_0^1 dx \int_0^1 f(x, y) dy = \frac{\pi}{4},$$

$$I_2 = \int_0^1 dy \int_0^1 f(x, y) dx, \text{ 由于 } \frac{\partial}{\partial x} \left(\frac{-x}{x^2 + y^2} \right) = f(x, y),$$

$$\text{故 } \int_0^1 f(x, y) dx = \int_0^1 \frac{\partial}{\partial x} \left(\frac{-x}{x^2 + y^2} \right) dx = \frac{-x}{x^2 + y^2} \Big|_0^1 = \frac{-1}{1 + y^2};$$

$$\text{所以 } I_2 = \int_0^1 dy \int_0^1 f(x, y) dx = \int_0^1 \frac{-1}{1 + y^2} dy = -\arctan y \Big|_0^1 \\ = -\frac{\pi}{4}.$$

$$\int_0^1 dx \int_0^1 f(x, y) dy \neq \int_0^1 dy \int_0^1 f(x, y) dx.$$

定积分与积分变量的记法无关

例6 求证 $\int_0^a dx \int_0^x f(y) dy = \int_0^a (a-x) f(x) dx \quad (a > 0)$

提示 左边的累次积分中, $f(y)$ 是 y 的抽象函数, 不能具体计算. 所以, **先交换积分次序.**

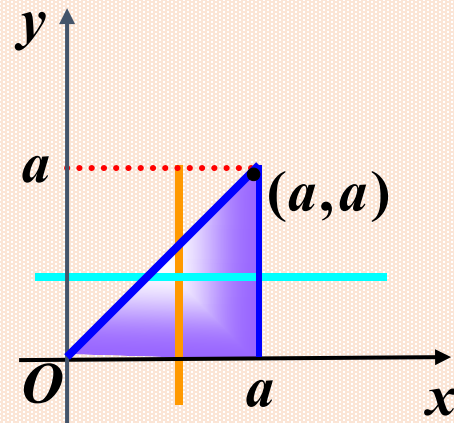
积分域 $0 \leq x \leq a, \quad 0 \leq y \leq x$

可表为 $0 \leq y \leq a, \quad y \leq x \leq a$

$$\int_0^a dx \int_0^x f(y) dy = \int_0^a dy \int_y^a f(y) dx$$

$$= \int_0^a f(y) \cdot x \Big|_y^a dy = \int_0^a (a-y) f(y) dy$$

$$= \int_0^a (a-x) f(x) dx \quad \text{证毕.}$$



例7 $\int_0^2 dx \int_x^2 e^{-y^2} dy = (\quad) - \frac{1}{2}(e^{-4} - 1)$

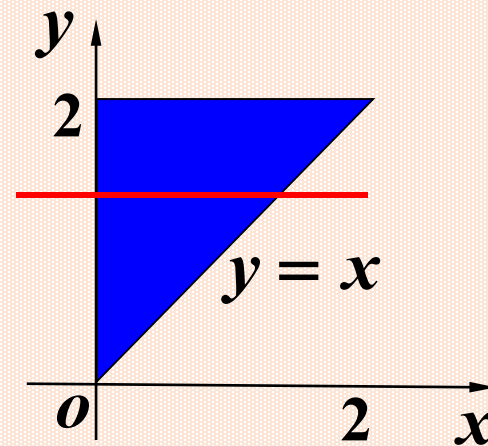
解 $\int_0^2 dx \int_x^2 e^{-y^2} dy$ 交换积分次序

$$= \int_0^2 dy \int_0^y e^{-y^2} dx$$

$$= \int_0^2 e^{-y^2} \cdot x \Big|_0^y dy$$

$$= \int_0^2 ye^{-y^2} dy$$

$$= -\frac{1}{2} \int_0^2 e^{-y^2} d(-y^2) = -\frac{1}{2}(e^{-4} - 1)$$



例8 计算积分

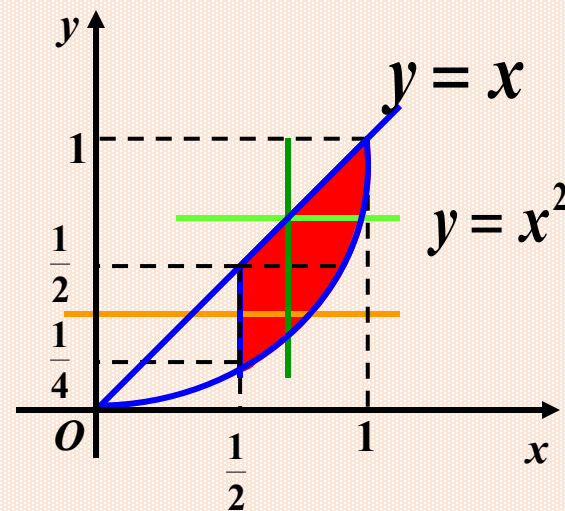
$$I = \int_{\frac{1}{4}}^{\frac{1}{2}} dy \int_{\frac{1}{2}}^{\sqrt{y}} e^{\frac{y}{x}} dx + \int_{\frac{1}{2}}^1 dy \int_y^{\sqrt{y}} e^{\frac{y}{x}} dx$$

解

$\because \int e^{\frac{y}{x}} dx$ 不能用初等函数表示,

$\therefore$ 先交换积分次序.

$$\begin{aligned} I &= \int_{\frac{1}{2}}^1 dx \int_{x^2}^x e^{\frac{y}{x}} dy \\ &= \int_{\frac{1}{2}}^1 x(e - e^x) dx \\ &= \frac{3}{8}e - \frac{1}{2}\sqrt{e} \end{aligned}$$



例9 计算 $\iint_D \frac{y \sin(x-1)}{x-1} dx dy, D: y^2 = x, y = x-2$

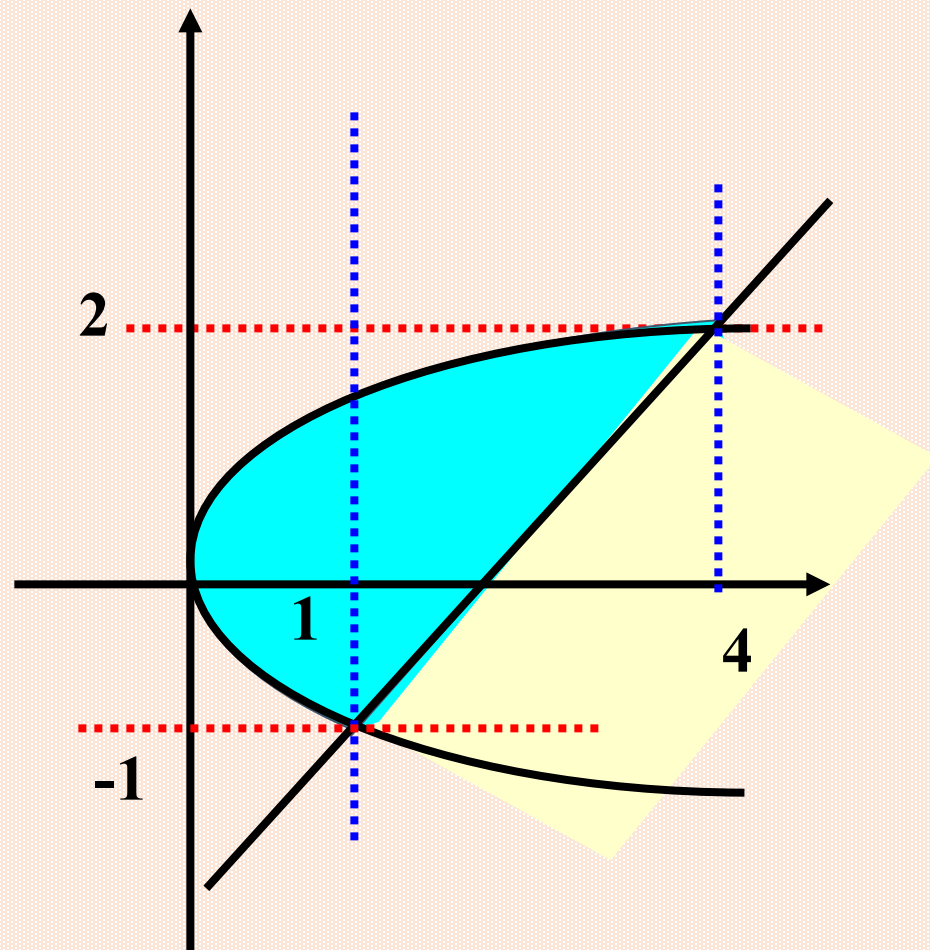
解 根据积分区域的特点

应先对 x 后对 y 积分

$$I = \int_{-1}^2 dy \int_{y^2}^{y+2} \frac{y \sin(x-1)}{x-1} dx$$

但由于 $\frac{\sin(x-1)}{x-1}$

对 x 的积分求不出，无法计算，须改变积分次序.



先 y 后 x 有

$$I = \int_0^1 dx \int_{-\sqrt{x}}^{\sqrt{x}} \frac{y \sin(x-1)}{x-1} dy + \int_1^4 dx \int_{x-2}^{\sqrt{x}} \frac{y \sin(x-1)}{x-1} dy$$

奇函数

$$= 0 + \int_1^4 \frac{1}{2} [x - (x-2)^2] \frac{\sin(x-1)}{x-1} dx$$

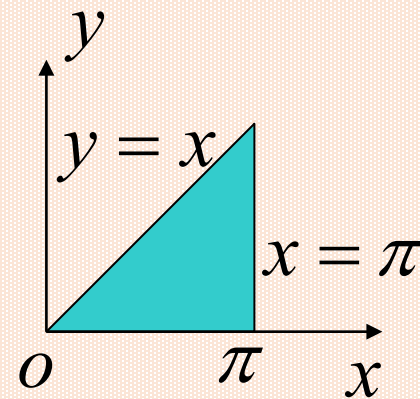
$$= \frac{1}{2} \int_1^4 (-x^2 + 5x - 4) \frac{\sin(x-1)}{x-1} dx$$

$$= \frac{1}{2} \int_1^4 (4-x) \sin(x-1) dx$$

$$= \frac{1}{2} (3 - \sin 3)$$

(1) D 是直线 $y = x$, $y = 0$, $x = \pi$ 所围成的闭区域.

则 $\iint_D \frac{\sin x}{x} dx dy =$ [填空1]



(2) $\int_0^1 dy \int_y^1 \frac{\tan x}{x} dx =$

(2017年高数二)

正常使用填空题需3.0以上版本雨课堂

作答

凡遇如下形式积分：

$$\int \frac{\sin x}{x} dx, \int \frac{\tan x}{x} dx, \int \sin x^2 dx, \int \cos x^2 dx,$$

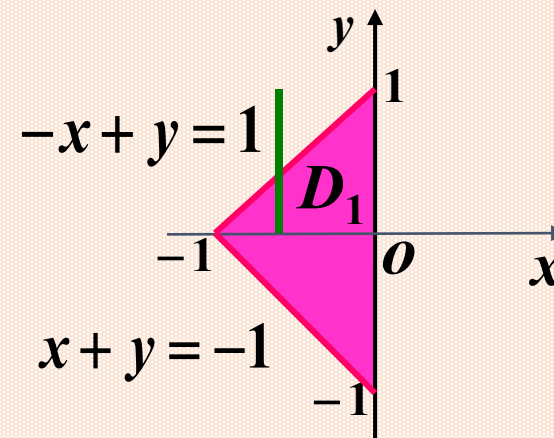
$$\int e^{-x^2} dx, \int e^{x^2} dx, \int e^{\frac{y}{x}} dx, \int \frac{dx}{\ln x}, \text{ 等等,}$$

一定要放在后面积分.

例10 计算 $\iint_D (|x| + |y|) dx dy$, $D: |x| + |y| \leq 1, x \leq 0$

解 积分区域 D 关于 x 轴对称,
被积函数关于 y 为偶函数.

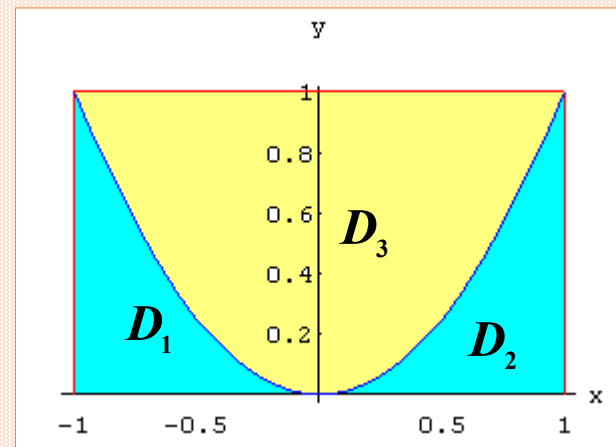
记 D_1 为 D 的 $y \geq 0$ 的部分. 则



$$\begin{aligned} \text{原式} &= 2 \iint_{D_1} (|x| + |y|) dx dy \\ &= 2 \iint_{D_1} (y - x) dx dy = 2 \int_{-1}^0 dx \int_0^{1+x} (y - x) dy \\ &= \frac{2}{3} \end{aligned}$$

例11 计算 $\iint_D |y - x^2| d\sigma$. 其中 $D: -1 \leq x \leq 1, 0 \leq y \leq 1$.

解 先去掉绝对值符号, 如图



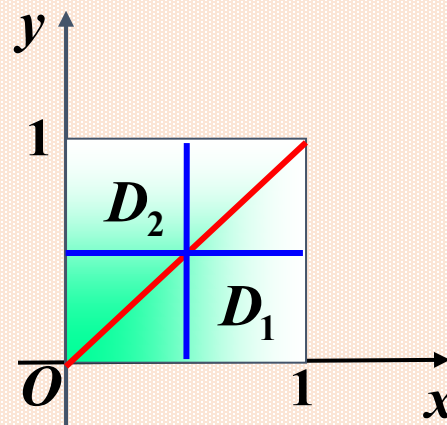
$$\begin{aligned} & \iint_D |y - x^2| d\sigma \\ &= \iint_{D_1+D_2} (x^2 - y) d\sigma + \iint_{D_3} (y - x^2) d\sigma \\ &= \int_{-1}^1 dx \int_0^{x^2} (x^2 - y) dy + \int_{-1}^1 dx \int_{x^2}^1 (y - x^2) dy = \frac{11}{15}. \end{aligned}$$

例12 计算二重积分 $\iint_D e^{\max\{x^2, y^2\}} dx dy$, 其中

$$D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}.$$

解 设 $D_1 = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq x\}$,

$$D_2 = \{(x, y) | 0 \leq x \leq 1, x \leq y \leq 1\},$$



$$\begin{aligned} \iint_D e^{\max\{x^2, y^2\}} dx dy &= \iint_{D_1} e^{\max\{x^2, y^2\}} dx dy + \iint_{D_2} e^{\max\{x^2, y^2\}} dx dy \\ &= \iint_{D_1} e^{x^2} dx dy + \iint_{D_2} e^{y^2} dx dy \end{aligned}$$

$$(\text{或} = 2 \int_0^1 dx \int_0^x e^{x^2} dy) = \int_0^1 dx \int_0^x e^{x^2} dy + \int_0^1 dy \int_0^y e^{y^2} dx = e - 1.$$

练习 设 $D = \{(x, y) | 0 \leq x \leq \sqrt{\pi}, 0 \leq y \leq \sqrt{\pi}\}$, 计算二重积分 $\iint_D \sin(\max\{x^2, y^2\}) dx dy$.
(南开大学2022)

二重积分的计算规律(2)

- (1) 注意利用对称性质, 以便简化计算;
- (2) 被积函数中含有绝对值、 $\max$ 、 $\min$ 符号时, 应将积分域分割成几个子域, 使被积函数在每个子域中保持同一表达式, 以消除被积函数中的绝对值、 $\max$ 、 $\min$ 等符号.

例13 求两个底圆半径为 R 的立体顶部 $x^2 + z^2 = R^2$ 及 立体底部 $x^2 + y^2 = R^2$ 方程分别为 $x^2 + y^2 = R^2$ 及 $x^2 + z^2 = R^2$ 所围成的立体的体积.

曲顶 $z = \sqrt{R^2 - x^2}$

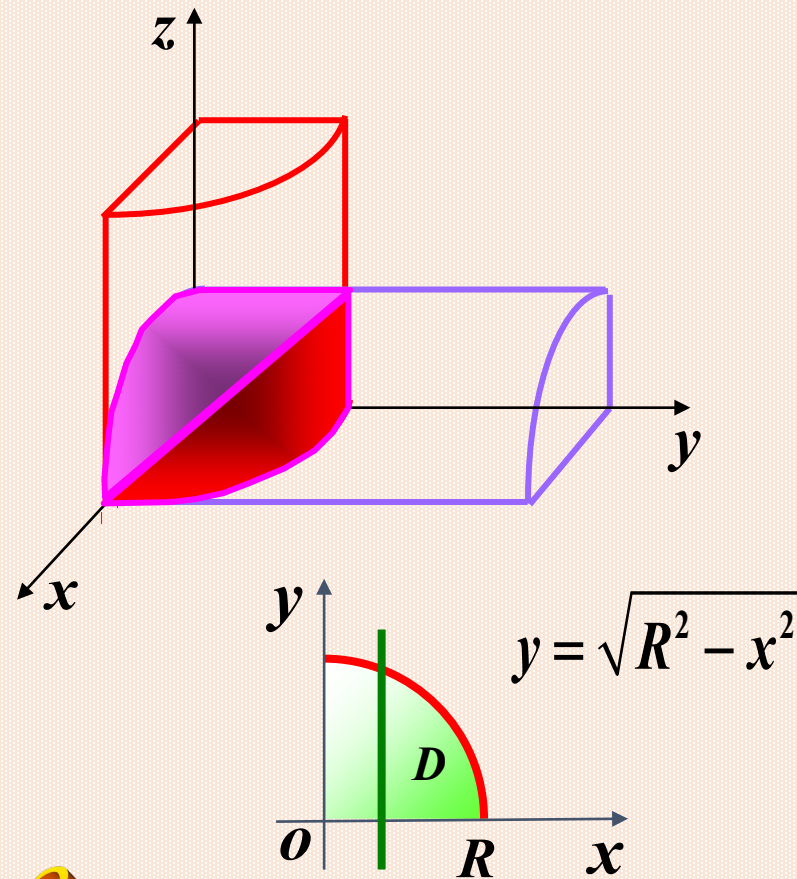
解 $V_1 = \iint_D f(x, y) d\sigma$

$$= \iint_D \sqrt{R^2 - x^2} d\sigma$$

$$= \int_0^R dx \int_0^{\sqrt{R^2 - x^2}} \sqrt{R^2 - x^2} dy$$

$$= \frac{2}{3} R^3$$

$V = 8V_1 = \frac{16}{3} R^3$ 还有别的做法吗 🧐



例14 设区域 $D = \{(x, y) | x^2 + y^2 \leq 1, y \geq 0\}$,

$$f(x, y) = y\sqrt{1-x^2} + x \iint_D f(x, y) dx dy, \text{ 求 } \iint_D xf(x, y) dx dy.$$

解 设 $\iint_D f(x, y) dx dy = A$, (2020年考研高数三)

依题意有 $f(x, y) = y\sqrt{1-x^2} + Ax$

两边同时积分有 $A = \iint_D f(x, y) dx dy = \iint_D y\sqrt{1-x^2} dx dy + \iint_D Ax dx dy$,

由于 D 关于 y 轴对称, 所以 $\iint_D Ax dx dy = 0$. 且有,

$$\begin{aligned} A &= 2 \iint_{D_1} y\sqrt{1-x^2} dx dy = 2 \int_0^1 \sqrt{1-x^2} dx \int_0^{\sqrt{1-x^2}} y dy \\ &= \int_0^1 (1-x^2)^{\frac{3}{2}} dx \stackrel{x=\sin t}{=} \int_0^{\frac{\pi}{2}} \cos^4 t dt = \frac{3\pi}{16}, \end{aligned}$$

$$\text{于是, } f(x, y) = y\sqrt{1-x^2} + \frac{3\pi}{16}x,$$

$$\iint_D xf(x, y)dxdy = \iint_D x\left(y\sqrt{1-x^2} + \frac{3\pi}{16}x\right)dxdy$$

$$\begin{aligned} & \xrightarrow{\text{对称性}} \frac{3\pi}{8} \iint_{D_1} x^2 dxdy \xrightarrow{\text{轮换对称性}} \frac{3\pi}{8} \iint_{D_1} y^2 dxdy \\ & = \frac{3\pi}{16} \iint_{D_1} (x^2 + y^2) dxdy = \frac{3\pi}{16} \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \cdot r dr = \frac{3\pi}{16} \cdot \frac{\pi}{2} \cdot \frac{1}{4} = \frac{3\pi}{128}. \end{aligned}$$

练习 设 $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$, 连续函数 $f(x, y)$ 满足

$$f(x, y) = e^{x^2+y^2} + xy \iint_{D_1} xyf(x, y)dxdy, \quad \text{求} \iint_{D_1} f(x, y)dxdy,$$

$$\text{其中 } D_1 = \{(x, y) | x^2 + y^2 \leq 1, x > 0, y > 0\}. \quad \frac{\pi}{4}(e-1) + \frac{9}{256}(e-1)^2$$